# Bayesian Probability

## Problem 1: The Unreliable Sensor

You are monitoring a machine component whose health parameter, $X$, is unknown. Based on past data, the prior belief about the health $X$ is modeled by the following probability density function (PDF):

$$f_X(x) = \begin{cases} 2x & \text{for } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

You use a sensor to measure the machine's performance, resulting in a reading $Y$. The sensor's reading is constrained by the machine's current health. If the health is $x$, the sensor reading $Y$ is uniformly distributed between 0 and $x$:

$$f_{Y|X}(y|x) = \begin{cases} \frac{1}{x} & \text{for } 0 < y < x \\ 0 & \text{otherwise} \end{cases}$$

**Question:** You observe a specific sensor reading of $Y = 0.5$. Calculate the **posterior probability density function** of the machine's health, $f_{X|Y}(x \mid y = 0.5)$.

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## Problem 2: The Uncertain Lightbulb

You are testing the lifetime of a new type of lightbulb. Let $Y$ be the time (in years) until the bulb burns out. The lifetime depends on an internal decay rate, $X$, which varies due to manufacturing differences.

- **The Prior (Manufacturing Rate):** $X$ follows an Exponential distribution with a mean of 1.

$$f_X(x) = e^{-x} \quad \text{for } x > 0$$

- **The Likelihood (Lifetime):** Given a specific decay rate $x$, the bulb's lifetime $Y$ follows an Exponential distribution with rate $x$.

$$f_{Y|X}(y|x) = xe^{-xy} \quad \text{for } y > 0$$

**Tasks:**

1. Find the **Joint Density** $f_{X,Y}(x, y)$ and specify its bounds.
2. Find the **Marginal Density** of the lifetime $f_Y(y)$.
3. You observe a bulb that lasts exactly $Y = 2$ years. Find the **Posterior Density** $f_{X|Y}(x \mid y = 2)$.